

STACK

Q10) Postfix to Prefix Conversion

DS	Session	Stack	Question Information	Level 1	Challenge 50
Question description					
Hassan gets a job in a software company in Hyderabad. The training period for the first three months is 20000 salary. Then incremented to 25000 salaries. Training is great but they will give you a programming task every day in three months. Hassan must finish it in the allotted time. His teammate Jocelyn gives him a task to complete the concept of Postfix to Prefix Conversion for a given expression. can you help him?					
Functional Description:					
- Read the Prefix expression in reverse order (from right to left) - If the symbol is an operand, then push it onto the Stack - If the symbol is an operator, then pop two operands from the Stack Create a string by concatenating the two operands and the operator after them. string = operand1 + operand2 + operator And push the resultant string back to Stack - Repeat the above steps until end of Prefix expression.					
Constraints					
the input should be a expressions					
Input Format					
Single line represents the postfix expressions					
Output Format					
Single line represents the prefix expression					

CODE:

```

C main.c > main
1  #include <stdio.h>
2  #include <string.h>
3  #include <ctype.h>
4
5  int isOperator(char ch) {
6      return (ch == '+' || ch == '-' || ch == '*' || ch == '/' || ch == '^');
7  }
8
9  int main(void) {
10     char exp[1005];
11     if (scanf(format: "%s", exp) != 1) return 0;
12
13     char stack[1005][1005];
14     int top = -1;
15
16     for (int i = 0; exp[i] != '\0'; i++) {
17         char ch = exp[i];
18
19         if (isalnum(ch)) {
20             char op[2] = {[0]=ch, [1]='\0'};
21             strcpy(dest: stack[++top], src: op);
22         } else if (isOperator(ch)) {
23             char op2[1005], op1[1005];
24             strcpy(dest: op2, src: stack[top--]);
25             strcpy(dest: op1, src: stack[top--]);
26
27             char temp[1005];
28             snprintf(s: temp, maxlen: sizeof(temp), format: "%c%s%s", ch, op1, op2);
29             strcpy(dest: stack[++top], src: temp);
30         }
31     }
32
33     if (top >= 0) {
34         printf(format: "%s\n", stack[top]);
35     }
36
37     return 0;
38 }

```

OUTPUT:

```

[suraj@suraj-83gs test]$ ./main
AB+CD-*
*+AB-CD
[suraj@suraj-83gs test]$

```